

# ADVANCED NETWORK SECURITY AND ARCHITECTURES

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Project Report [EN]

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# 1 Introduction

Group Encrypted Transport VPN (GETVPN) is a Cisco proprietary technology designed to provide scalable, any-to-any encrypted communication over private WAN infrastructures such as MPLS networks. Unlike traditional point-to-point IPsec VPNs, which require a dedicated tunnel between each pair of sites, GETVPN operates on a group key model: all participating routers share the same encryption keys and policy, allowing any member to securely communicate with any other member without the overhead of per-pair tunnel establishment.

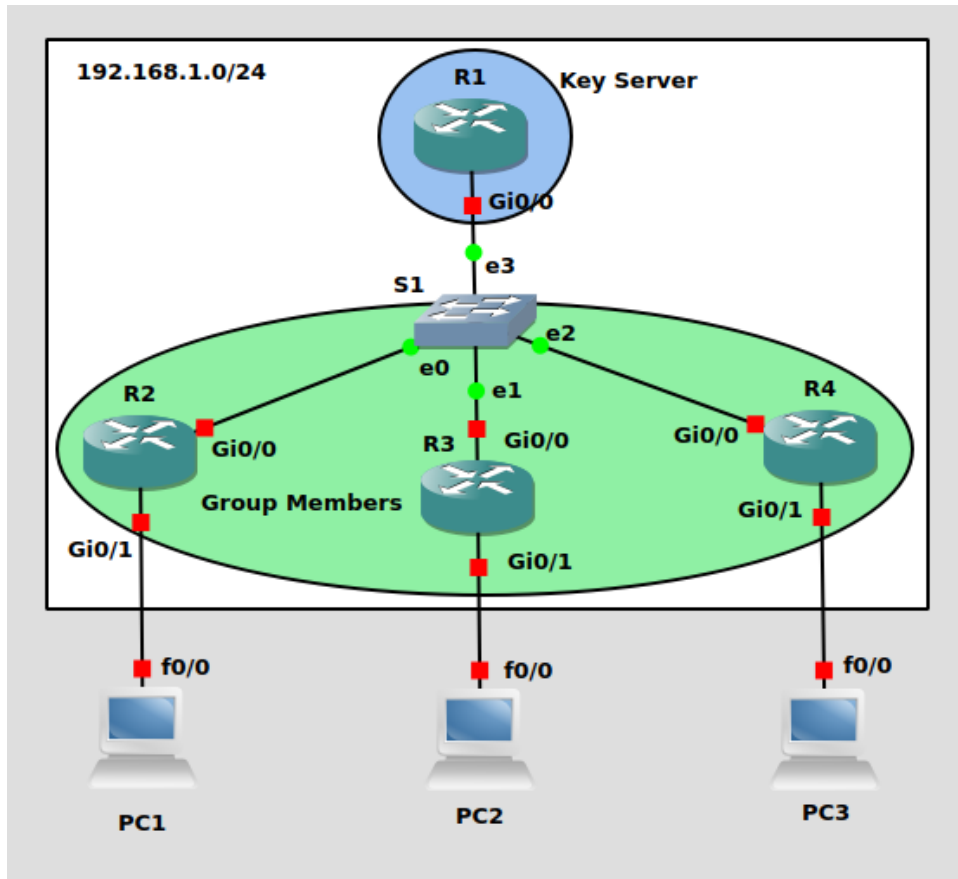
The GETVPN architecture is built around two roles. The **Key Server (KS)** is responsible for generating and distributing cryptographic keys and security policies to all group participants. The **Group Members (GMs)** are the routers that register with the Key Server, receive the shared keys and traffic policies, and then encrypt or decrypt data traffic autonomously. Once registered, GMs communicate directly with each other; the Key Server is not involved in the data plane.

The goal of this project is to deploy and study a GETVPN solution in a simulated environment using GNS3, analyse the key management protocol exchanges, observe traffic encryption behaviour, and extend the base configuration with additional features. The remainder of this report describes the topology, the configuration steps, the observations made during the lab tasks, and the extensions implemented.

# 2 Topology and Base Configuration

The network topology used in this project is shown in Figure 1. It consists of four Cisco IOSv routers interconnected through a central switch. Router R1 acts as the Key Server and is connected to the shared segment 192.168.1.0/24 with address 192.168.1.254. Routers R2, R3, and R4 are the Group Members, with addresses 192.168.1.2, 192.168.1.3, and 192.168.1.4 respectively on their gi0/0 interfaces. Each Group Member also has a LAN-facing interface (gi0/1) connecting to a dedicated subnet: R2 serves 192.168.10.0/24, R3 serves 192.168.20.0/24, and R4 serves 192.168.30.0/24.

Three end hosts (PC1, PC2, PC3) are attached to these LAN subnets. Since the lab environment does not include dedicated PC images, Cisco 3725 routers were used to simulate the hosts; IP routing was disabled on these devices (`no ip routing`), and a default gateway static route was configured on each to point to the respective Group Member. Each simulated PC was assigned the .100 address of its subnet (e.g., PC1 at 192.168.10.100) and its icon was changed to a PC in GNS3 for clarity.



**Figure 1:** Project topology. R1 is the Key Server; R2, R3, and R4 are Group Members.

OSPF was chosen as the routing protocol and was configured on all four routers to advertise all connected networks. Prior to applying any GETVPN or IPsec configuration, end-to-end reachability was verified by pinging between all three PCs and between all router interfaces. All pings were successful, confirming that the routing layer was fully operational before any security configuration was introduced.

## 3 GETVPN Configuration

### 3.1 Overview of the Configuration Model

GETVPN security configuration is divided between the Key Server and the Group Members. The Key Server defines all security policy — the IKE parameters, the cryptographic algorithms, the traffic selector (which packets to encrypt), and the keys themselves — and distributes this policy to GMs during their registration. GMs only need to know how to authenticate with the KS and where to register; they receive everything else from the KS automatically.

The key management protocol used to distribute keys and policy is **GDOI (Group Domain of Interpretation)**, which runs over UDP port 848. The registration process has two phases: an IKE Phase 1 exchange that establishes a secure authenticated channel between the GM and the KS, followed by a **GROUPKEY-PULL** exchange in which the GM requests and receives the group keys and IPsec policy.

## 3.2 Key Server Configuration (R1)

### 3.2.1 IKE Phase 1 Policy

The first step is to define the parameters for the IKE Phase 1 exchange that will protect the GDOI registration channel between each GM and the KS. AES encryption, SHA hashing, pre-shared key authentication, and Diffie-Hellman group 5 are used. The wildcard address 0.0.0.0 allows the KS to accept registrations from any GM without listing each one individually.

```
! IKE Phase 1
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config)#crypto isakmp key saar address 0.0.0.0
```

### 3.2.2 Transform Set and IPSec Profile

The *transform set* defines the cryptographic algorithms that will be applied to data traffic: ESP with AES for encryption and SHA-HMAC for integrity. The *IPSec profile* bundles the transform set into a named object that is later referenced inside the GDOI group configuration.

```
! Transform set (cipher suite for data traffic)
R1(config)#crypto ipsec transform-set myTSet esp-aes esp-sha-hmac

! IPSec profile (references the transform set)
R1(config)#crypto ipsec profile myIPSecProfile
R1(ipsec-profile)#set transform-set myTSet
```

### 3.2.3 RSA Key Pair

The KS signs its rekey messages using an RSA key pair so that GMs can verify the authenticity and integrity of key updates. The label `myRSAKeys` is used to reference this key pair in the GDOI group configuration.

```
R1(config)#crypto key generate rsa modulus 1024 label myRSAKeys
```

### 3.2.4 Traffic Selector ACL

GETVPN does not encrypt all traffic indiscriminately. An extended ACL defines the *interesting traffic* — the flows that should be protected. This ACL is configured on the KS and pushed to all GMs during registration. In this lab, only ICMP traffic is protected, which is sufficient for the ping-based verification tasks.

```
! Interesting traffic (what gets encrypted)
R1(config)#ip access-list extended ICMP
R1(config-ext-nacl)#permit icmp any any
```

### 3.2.5 GDOI Group

The GDOI group is the central object of the KS configuration. It ties together the identity number (which GMs must match to join the group), the KS address, the rekey mechanism, and the IPsec SA policy including the traffic selector.

```
R1(config)#crypto gdoi group myGDOIGroup
R1(config-gkm-group)#identity number 123
R1(config-gkm-group)#server local
R1(gkm-local-server)#address ipv4 192.168.1.254
R1(gkm-local-server)#rekey authentication mypubkey rsa myRSAKeys
R1(gkm-local-server)#rekey transport unicast
R1(gkm-local-server)#sa ipsec 10
R1(gkm-sa-ipsec)#profile myIPSecProfile
R1(gkm-sa-ipsec)#match address ipv4 ICMP
```

The `rekey transport unicast` directive instructs the KS to send key refresh messages directly to each registered GM individually, as opposed to multicast delivery.

## 3.3 Group Member Configuration (R2, R3, R4)

The configuration is identical on all three GMs; only the hostname differs. Each GM configures the same IKE Phase 1 parameters and pre-shared key as the KS, joins the GDOI group by its identity number and the KS address, and applies a crypto map to its WAN-facing interface to activate the received policy.

```
! IKE Phase 1 (must match KS policy)
Rx(config)#crypto isakmp policy 10
Rx(config-isakmp)#encryption aes
Rx(config-isakmp)#hash sha
Rx(config-isakmp)#authentication pre-share
Rx(config-isakmp)#group 5
Rx(config)#crypto isakmp key saar address 192.168.1.254

! Transform set (required locally, actual policy comes from KS)
Rx(config)#crypto ipsec transform-set TRANSFORM_SET esp-aes esp-sha-hmac

! GDOI group (must match KS identity number)
Rx(config)#crypto gdoi group myGDOIGroup
Rx(config-gkm-group)#identity number 123
Rx(config-gkm-group)#server address ipv4 192.168.1.254

! Crypto map applied to WAN interface
Rx(config)#crypto map myCryptoMap 10 gdoi
Rx(config-crypto-map)#set group myGDOIGroup
Rx(config)#interface gi0/0
Rx(config-if)#crypto map myCryptoMap
```

It is worth noting that the transform set defined locally on a GM is not used for data traffic; the actual IPSec policy (including transform set, traffic selector, and keys) is always received from the KS. The local transform set is only relevant for the IKE Phase 1 exchange during registration.

## 4 Verification

### 4.1 Key Server Status

After all GMs were configured, the `show crypto gdoi` and `show crypto gdoi ks members` commands were used on R1 to confirm correct operation.

```
R1#show crypto gdoi
GROUP INFORMATION

  Group Name           : myGDOIGroup (Unicast)
  Re-auth on new CRL   : Disabled
  Group Identity       : 123
  Group Type           : GDOI (ISAKMP)
  Crypto Path          : ipv4
  Key Management Path   : ipv4
  Group Members        : 3
  IPSec SA Direction   : Both
  IP D3P Window        : Disabled
  CKM status           : Disabled
  Group Rekey Lifetime : 86400 secs
  Group Rekey
    Remaining Lifetime  : 86355 secs
    Time to Rekey       : 86130 secs
    Acknowledgement Cfg : Cisco
  Rekey Retransmit Period : 10 secs
  Rekey Retransmit Attempts: 2
  Group Retransmit
    Remaining Lifetime  : 0 secs

  IPSec SA Number      : 10
  IPSec SA Rekey Lifetime: 3600 secs
  Profile Name         : myIPSecProfile
  Replay method        : Count Based
  Replay Window Size    : 64
  Tagging method       : Disabled
  SA Rekey
    Remaining Lifetime  : 3556 secs
    Time to Rekey       : 3170 secs
  ACL Configured       : access-list ICMP

  Group Server list    : Local
```

**Figure 2:** show crypto gdoi on the Key Server (R1). Three Group Members registered.



```

R1#show crypto gdoi ks members

Group Member Information :

Number of rekeys sent for group myGDOIGroup : 0
Number of retransmits during the last rekey for group myGDOIGroup : 0
Duration of the last rekey for group myGDOIGroup : 0 msec

Group Member ID   : 192.168.1.2           GM Version: 1.0.17
Group ID          : 123
Group Name        : myGDOIGroup
Group Type        : GDOI (ISAKMP)
GM State          : Registered
Key Server ID     : 192.168.1.254
Rekeys sent       : 0
Rekeys retries    : 0
Rekey Acks Rcvd   : 0
Rekey Acks missed : 0

Sent seq num : 0      0      0      0
Rcvd seq num : 0      0      0      0

Group Member ID   : 192.168.1.3           GM Version: 1.0.17
Group ID          : 123

Group Member Information :

Group Name        : myGDOIGroup
Group Type        : GDOI (ISAKMP)
GM State          : Registered
Key Server ID     : 192.168.1.254
Rekeys sent       : 0
Rekeys retries    : 0
Rekey Acks Rcvd   : 0
Rekey Acks missed : 0

Sent seq num : 0      0      0      0
Rcvd seq num : 0      0      0      0

Group Member ID   : 192.168.1.4           GM Version: 1.0.17
Group ID          : 123
Group Name        : myGDOIGroup
Group Type        : GDOI (ISAKMP)
GM State          : Registered
Key Server ID     : 192.168.1.254
Rekeys sent       : 0
Rekeys retries    : 0
Rekey Acks Rcvd   : 0
Rekey Acks missed : 0

Group Member Information :

Sent seq num : 0      0      0      0
Rcvd seq num : 0      0      0      0

```

**Figure 3:** show crypto gdoi ks members on R1. All three GMs (192.168.1.2, .3, .4) show GM State: Registered.

The output confirms that all three GMs successfully completed the GROUPKEY-PULL exchange. Each entry shows GM State: Registered and identifies the correct Key Server address (192.168.1.254). The rekey counters are all zero, which is expected since no TEK refresh has occurred yet.

## 4.2 Group Member Status

The `show crypto gdoi` and `show crypto gdoi ipsec sa` commands were run on R2 as a representative Group Member.

Several important aspects of GETVPN operation are visible in the output from Figure 4:

- **Registration status:** R2 shows `Registration status: Registered`, confirming it successfully joined the group.
- **ACL received from KS:** The traffic selector (`permit icmp any any`) was pushed by the KS and installed on R2 as `gdoi_group.myGDOIGroup_temp_acl`. GMs do not define this locally.
- **TEK policy:** The Traffic Encryption Key policy shows `transform:esp-aes esp-sha-hmac` and SPI `0x6D2F03B9`, with a remaining lifetime of 3478 seconds. This is the key used to encrypt ICMP traffic between GMs.
- **KEK policy:** The Key Encryption Key uses 3DES with a lifetime of 86276 seconds and is used to protect future rekey messages from the KS. Notably, the KEK uses 3DES regardless of the AES algorithm specified in the IKE policy; this is a known behaviour in the IOS version used.
- **Active TEK Number: 1:** Confirms one TEK is active and ready to encrypt matching traffic.

```

R2#show crypto gdoi
GROUP INFORMATION

  Group Name           : myGDOIGroup
  Group Identity       : 123
  Group Type           : GDOI (ISAKMP)
  Crypto Path          : ipv4
  Key Management Path  : ipv4
  Rekeys received      : 0
  IPSec SA Direction  : Both

  Group Server list    : 192.168.1.254

Group Member Information For Group myGDOIGroup:
  IPSec SA Direction   : Both
  ACL Received From KS : gdoi_group_myGDOIGroup_temp_acl

  Group member         : 192.168.1.2      vrf: None
    Local addr/port    : 192.168.1.2/848
    Remote addr/port   : 192.168.1.254/848
    fvrf/ivrf         : None/None
    Version            : 1.0.17
    Registration status : Registered
    Registered with    : 192.168.1.254
    Re-registers in    : 3229 sec
    Succeeded registration: 1
    Attempted registration: 1
    Last rekey from    : 0.0.0.0
    Last rekey seq num : 0
    Unicast rekey received: 0
    Rekey ACKs sent    : 0
    Rekey Received     : never
    DP Error Monitoring : OFF
    IPSEC init reg executed : 0
    IPSEC init reg postponed : 0
    Active TEK Number  : 1
    SA Track (OID/status) : disabled

  Rekeys cumulative
    Total received      : 0
    After latest register : 0
    Rekey Acks sents    : 0

  ACL Downloaded From KS 192.168.1.254:
  access-list permit icmp any any

KEK POLICY:
  Rekey Transport Type : Unicast
  Lifetime (secs)      : 86276
  Encrypt Algorithm     : 3DES
  Key Size              : 192
  Sig Hash Algorithm    : HMAC_AUTH_SHA
  Sig Key Length (bits) : 1296

TEK POLICY for the current KS-Policy ACES Downloaded:
GigabitEthernet0/0:
  IPsec SA:
    spi: 0x6D2F03B9(1831797689)
    KGS: Disabled
    transform: esp-aes esp-sha-hmac
    sa timing:remaining key lifetime (sec): (3478)
    Anti-Replay(Counter Based) : 64
    tag method : disabled
    alg key size: 16 (bytes)
    sig key size: 20 (bytes)
    encaps: ENCAPS_TUNNEL

```

**Figure 4:** show crypto gdoi on R2. Registration confirmed; TEK and KEK policies downloaded from KS.

## **5 Lab Tasks**

### **5.1 IKE and GROUPKEY-PULL Analysis (Wireshark)**

### **5.2 SPI Consistency Across Group Members**

### **5.3 TEK Rekey Observation**

## **6 GETVPN Extensions**

## **7 Conclusion**